Semantic Compasses in Attention Heads

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Abstract

Large language models have billions of parameters, and nobody fully understands what specific circuits are doing inside. This note walks through one concrete result: in each of four different model families (GPT-2, Phi-3, Gemma, Llama-3.2), there is a *single 1-D direction inside one attention head* that encodes **gender**. Zeroing that direction removes gender bias on WinoGender at near-zero fluency cost; over-amplifying it reverses bias. We explain what that means, how we found it, and where it works (and does not). All claims are backed by code, CSV outputs, and the five figures embedded below.

1 What we set out to do

Large language models (GPT-2, Phi-3, Gemma, ...) are black boxes. We wanted to find concrete, surgical, human-meaningful *dials* inside them — small pieces that encode a single concept such as *gender* — and prove three things:

1. they exist,
2. they are **causal** (the model really uses them), and
3. we can turn them up or down to change behaviour.

2 Background: what is inside a transformer

A transformer is a stack of **layers** (GPT-2 has 12, Phi-3 has 32, Llama-3.2-3B has 28). Each layer has many **attention heads** (GPT-2: 12 per layer, 144 total; Llama-3.2-3B: 24 per layer, 672 total). Each head reads from the residual stream — the model's running "working memory" — and writes something back.

Each head's write behaviour is captured by a matrix W_{OV} (Output-Value). SVD of W_{OV},

$$W_{OV} = U \Sigma V^\top,$$

gives an ordered basis for what the head can write: the first row of $V^\top$ is the head's strongest write direction, the second is the next strongest, and so on. We call each such row a **candidate compass axis**.

Technical: how W_{OV} is computed

For attention head h in layer ℓ, TransformerLens exposes `model.W_V[ℓ, h]` ($\mathbb{R}^{d_{\text{model}} \times d_h}$) and `model.W_O[ℓ, h]` ($\mathbb{R}^{d_h \times d_{\text{model}}}$). We form the composite $W_{OV} = W_V W_O$ and run `torch.linalg.svd(W_OV, full_matrices=False)`. *Rows* of $V^\top$ (i.e. `Vt[k]`) are the candidate compass axes, ordered by decreasing singular value σ_k.

3 Step 1 — the scan: does any head have a compass?

For every `(layer, head, 2-D plane of top SVs)` we asked: if we inject a rotating 2-D signal into that plane and observe what the model outputs, does the output also rotate around smoothly — like a compass needle?

What "the model outputs" means here. We don't use a labelled dataset for the scan. We use **3 short neutral continuation prompts** (*"The person said that"*, *"Then they said that"*, *"Afterwards, the speaker said that"*) and a **token-probe pair**: the logit gap between `" he"` and `" she"`. The scan asks: as we rotate the injected signal through 360°, does that gap trace a clean cosine? No WinoGender sentences, no BLS labels — just a probe along the gender axis.

Two tests had to pass:

- **Amplitude linearity** — doubling input magnitude doubles output magnitude ($R^2 > 0.95$).
- **Phase stability** — output angle stays consistent ($|\Delta\varphi| < 10°$).

Technical: compass test

At the last-token position of each of the 3 neutral prompts we inject $z(\theta, \alpha) = \alpha(\sigma_i \cos\theta \cdot u_i + \sigma_j \sin\theta \cdot u_j)$ into `blocks.`ℓ`.hook_resid_pre` for $\theta \in \{0, 30°, \ldots, 330°\}$ (12 angles) and $\alpha \in \{3, 10\}$. Readout: $LD(\theta, \alpha) = \text{logit}_{\text{" he"}} - \text{logit}_{\text{" she"}}$. We fit $LD(\theta) = \mu + A\cos(\theta - \varphi)$ at each α. Plane passes iff (i) linearity of A in α gives $R^2 > 0.95$, (ii) amplitude slope $A/\alpha > 0.2$, and (iii) phase drift $|\varphi(\alpha_{\text{hi}}) - \varphi(\alpha_{\text{lo}})| < 10°$.

Finding

Passing rates: **GPT-2: 16/864** (1.9%)**, Phi-3: 14/1344** (1.0%)**, Gemma: 3/240** (1.2%)**, Llama-3.2-3B: 9/4032** (0.2%). Compasses are rare; most heads fail both tests. Llama-3B is $\sim 5\times$ rarer than the others, suggesting its attention heads carry more compositional (less axis-aligned) features — yet the few compasses it does have are the *sharpest* we have seen (amp_slope up to 0.94).

4 Step 2 — are compasses meaningful, or random coincidences?

About 2% of planes pass the scan. Two things could explain that:

- **(A) The scan is finding real structure** — gender really lives in those planes.
- **(B) The scan is fooling itself** — any random 2-D plane inside an attention head looks compass-like, so 2% is just the base rate.

To decide between (A) and (B), we run the *exact same* test on **random 2-D planes** and compare pass rates. This baseline is called a *null* — "what would we see if nothing is there." If random planes pass $\sim 2\%$ too, our compasses are noise (option B). If random planes fail far more often, our compasses are real (option A).

The subtle question: *where do we draw the random planes from?* Two honest choices:

1. **Loose null (full_ov):** random planes from the head's *whole* output space. Asks: *"is our compass better than any random direction the head could write?"*
2. **Strict null (top-4):** random planes from only the head's top-4 strongest singular directions — the same neighbourhood our compass comes from. Asks: *"is our compass better than its nearest alternatives?"*

What we found.

Null type	GPT-2	Phi-3	Gemma	Llama-3.2-3B	Verdict
Loose (full_ov)	2.06% vs 0.44%	1.18% vs 0.28%	2.02% vs 0%	—	Compass **wins**
Strict (top-4)	2.06% vs 2.51%	1.18% vs 1.62%	2.02% vs 1.55%	0.22% vs 0.15%	Compass **wins/ties**

Note on Llama-3B. Because Llama's compass rate is already tiny (0.22%) and the strict-null rate is smaller still (0.15%), we did not run the loose null for Llama: it is strictly easier than the strict null and would only confirm the separation. The strict-null comparison is conservative and already positive.

What the tie means. The top-4 subspace of these specific heads is *densely* compass-like — gender is not on a lone magic axis; it is spread across a small 2-D patch of directions near the top of the SVD. Our chosen axis is one clean member of that patch.

Finding

The honest claim is not "we found **the** gender axis" but **"gender is concentrated in a low-dim subspace of one attention head."** The loose null justifies *the head carries gender*; the strict-null tie explains *where inside the head it lives*.

5 Step 3 — what do these compasses actually encode?

We picked the strongest compass per model:

- GPT-2: **L10 H9** (SV0)
- Phi-3: **L28 H1** (SV0)
- Gemma-2: **L21 H4** (SV0)
- Llama-3.2-3B: **L26 H14** (**SV2**)

Decoding the compass direction through the unembedding matrix W_U (project the direction to logit space and read top tokens) gives **he, him, his, man** on one side and **she, her, hers, woman** on the other. On Phi-3 and Llama we also see cross-lingual alignment: *lui*, , *elle*, *ella*, , , — so it is language-agnostic *gender*, not just English pronouns.

Llama's SV quirk. Unlike the other three models, the gender axis in Llama-3.2-3B is *not* on the head's top singular vector; it sits at SV2. SV0 and SV1 decode to register/punctuation noise; SV2 cleanly encodes she/her// versus his//jego. This is consistent with Llama's heavier-tailed spectrum: σ_0 through σ_7 of L26 H14 capture 52.6% of the variance (vs 23–36% for the other models), and gender happens to sit slightly below the top two write directions rather than at the very top. The causal story and the α-sweep intervention apply identically.

Technical: decoding a direction

Given compass axis $u \in \mathbb{R}^{d_{\text{model}}}$, compute $\ell = W_U u$ and read the top-k highest and lowest entries of ℓ. Optionally centre by subtracting the mean unembedding row so generic high-frequency tokens do not dominate.

6 Step 4 — is it causal? (the key test)

Finding a pattern $\neq$ proving the model uses it. Our causal test is a *projection ablation*: at the compass head's layer we subtract the component of the residual along u,

$$r_{\text{new}} \;=\; r \;-\; \alpha \langle r, u \rangle \, u,$$

with $\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2\}$. $\alpha = 0$ does nothing, $\alpha = 1$ zeros the compass, $\alpha > 1$ over-shoots into the anti-compass direction.

We measure on **WinoGender** (720 sentences such as *"The nurse looked at **his** notes"*). Two bias metrics:

- **stereo_corr** — Pearson correlation between BLS real-world %-male per occupation and the model's mean ($\text{logit}_{he} - \text{logit}_{she}$). 0 = unbiased; +0.6 = strongly biased; −0.1 = reversed.
- **stereo_delta** — a simpler backup metric: mean logit gap on male-dominated occupations minus female-dominated.

Technical: hook installation

We install a forward hook at `blocks.ℓ.hook_resid_post`:

```
def _h(r, hook):
    proj = (r * u).sum(dim=-1, keepdim=True)
    return r - alpha * proj * u
```

and call `model.run_with_hooks(tokens, fwd_hooks=[(..., _h)])`.

Finding

Bias hits zero near α=1.5 and flips sign at α=2 in all four models. Both bias metrics agree on all 36 data points. *Fluency* on WikiText-2 moves by at most 1 PPL point; on Llama-3.2-3B the PPL cost at α=1.5 is just +0.035, the cleanest intervention of the four.

7 Step 5 — compare against a known baseline (ActAdd)

ActAdd (Turner et al. 2023) takes the difference of mean residuals on "he-prompts" vs "she-prompts" and subtracts it. It is the standard direction-based steering baseline.

Technical: ActAdd direction

We build the ActAdd direction from *paired minimal prompts* (*"He walked into the room and"* vs *"She walked into the room and"*; 12 pairs), capture the residual at the *last*-token position of `hook_resid_post` in the target layer, and take $d = \mu_{\text{he}} - \mu_{\text{she}}$. Edit is identical in shape to compass: $r_{\text{new}} = r - \alpha \langle r, \hat{d} \rangle \hat{d}$. *Crucially* ActAdd is built on prompts with no WinoGender overlap — otherwise the direction would leak the eval.

Finding

Compass at α=1.5 beats ActAdd at α=1.0 on WinoGender in all four models and costs 4–20× less perplexity. Compass is *one* specific 1-D axis (targeted); ActAdd is a whole-layer direction (diffuse). On Llama-3.2-3B the gap is particularly large: compass moves `stereo_corr` from 0.73 to −0.18, while ActAdd barely moves it (0.73 → 0.72).

8 Step 6 — cherry-pick defense

Reviewer worry: "you picked the single best head after seeing results." Counter-test: apply the same α=1 edit to the *other* passing compass heads in each model.

- GPT-2: L9 H7 and L11 H1 are also compasses, but they barely move WinoGender bias (0.603 → 0.616/0.592).
- Phi-3: L27 H16 and L24 H10 are compasses, but they barely move bias either (0.638 → 0.584/0.645).

Finding

This is a *dissociation*: secondary compass heads carry pronoun-salience but *not* the occupation-to-gender mapping. The primary head is specifically the *occupational-stereotype readout*. That makes the finding richer, not weaker.

9 Step 6.5 — a taxonomy of pronoun compasses

The head-specificity result above raises the obvious question: if the secondary heads pass the compass scan but do *not* debias, what do they actually encode? We decoded the top-8 singular-vector axes of every passing compass head through the unembedding W_U (per-row mean-centred) and read the top tokens on each side.

Technical: axis selection

For each head we compute $W_{OV} = W_V W_O$, take the compact SVD, and form the cumulative variance ratio $c_K = \sum_{k \le K} \sigma_k^2 / \sum_k \sigma_k^2$. We decode at least the top-4 axes (the basis the scan used) and extend to at most the top-8, stopping once $c_K \ge 0.90$. In practice no head reaches 0.90 within 8 axes: the W_{OV} spectra of these heads are heavy-tailed, with $c_8 \in [0.23, 0.36]$. This is itself a finding — the compass direction is a small *slice* of a broadly-writing head, not a dominant mode.

Reading the SV0 axis of every passing head gives a taxonomy:

Model	Head (axis)	top tokens $+v$ / $-v$	Encodes	Debiases?
GPT-2	**L10 H9 (SV0)**	his, he / her, she	binary gender (en)	**yes**
GPT-2	L9 H7 (SV0)	their, they / his, he	number (3pl vs 3sg.masc)	no
GPT-2	L11 H1 (SV0)	(masc names) / widow, daughter	kinship/role gender	no
Phi-3	**L28 H1 (SV0)**	his/lui/ / her/elle/ella	multilingual gender	**yes**
Phi-3	L27 H16 (SV0)	him, lui / they, she	pronoun salience (mixed)	no
Phi-3	L24 H10 (SV0)	their, loro / its, him	number × animacy	no
Phi-3	L26 H22 (SV0)	this, dieses / these, estos	demonstrative number	no
Phi-3	L25 H24 (SV0)	their, themselves / your, our	person (3pl vs 1/2)	no
Gemma	**L21 H4 (SV0)**	he//él / she/	multilingual gender	**yes**
Llama-3.2	**L26 H14 (SV2)**	she/her// / his//jego	multilingual gender	**yes**
Llama-3.2	L20 H13 (SV0)	his / her,	binary gender (secondary)	(not tested)
Llama-3.2	L16 H20 (SV0)	their / his,	number (3pl vs 3sg.masc)	(not tested)

Finding

"Passing the compass scan" is *not* the same as "encoding gender." Of 12 heads that pass across 4 models, only 4 (one per model, highlighted above) carry the gender axis that actually debiases; the others are orthogonal pronoun-feature compasses — number, person, demonstrative, kinship. Llama adds a wrinkle: its gender axis sits at SV2 rather than SV0, because the head's top two write directions are register/punctuation noise. Across all four models the gender axis is *one* axis among several, not the dominant one. This is what makes the causal intervention sharp: ablating the gender compass does not wipe out pronoun grammar, because gender lives on a different axis than number and person.

Full decoded dictionaries (top-8 axes per head, top-15 tokens per side) are in `helix_usage_validated/compass_`

10 Step 7 — cross-benchmark check (StereoSet)

To avoid a "one-metric fluke" objection, we ran the same intervention on **StereoSet** — a completely different benchmark that scores the full-sentence log-probability of *stereotype*, *anti-stereotype* and *unrelated* completions.

Technical: StereoSet scoring

For each context with three candidate fillers and gold-labels $\{0, 1, 2\}$, we compute sum-log-prob of each candidate sentence under the model (optionally with hooks installed) and count, per pair, `Stereotype Score (SS)` = % where stereotype > anti, and `LMS` = % where either meaningful option beats the unrelated one. `ICAT` combines both: $\text{LMS} \times \min(\text{SS}, 100 - \text{SS})/50$.

Finding

WinoGender result replicates on all four models. StereoSet replicates on GPT-2 only; on Phi-3, Gemma, and Llama-3.2-3B the compass acts sharply at the pronoun slot but does not propagate into whole-sentence log-prob averages (SS stays within ± 0.5 of baseline at α=1.5 across those three models). That is **an honest boundary of the finding**: compass is a position-local, gender-specific direction, not a universal knob.

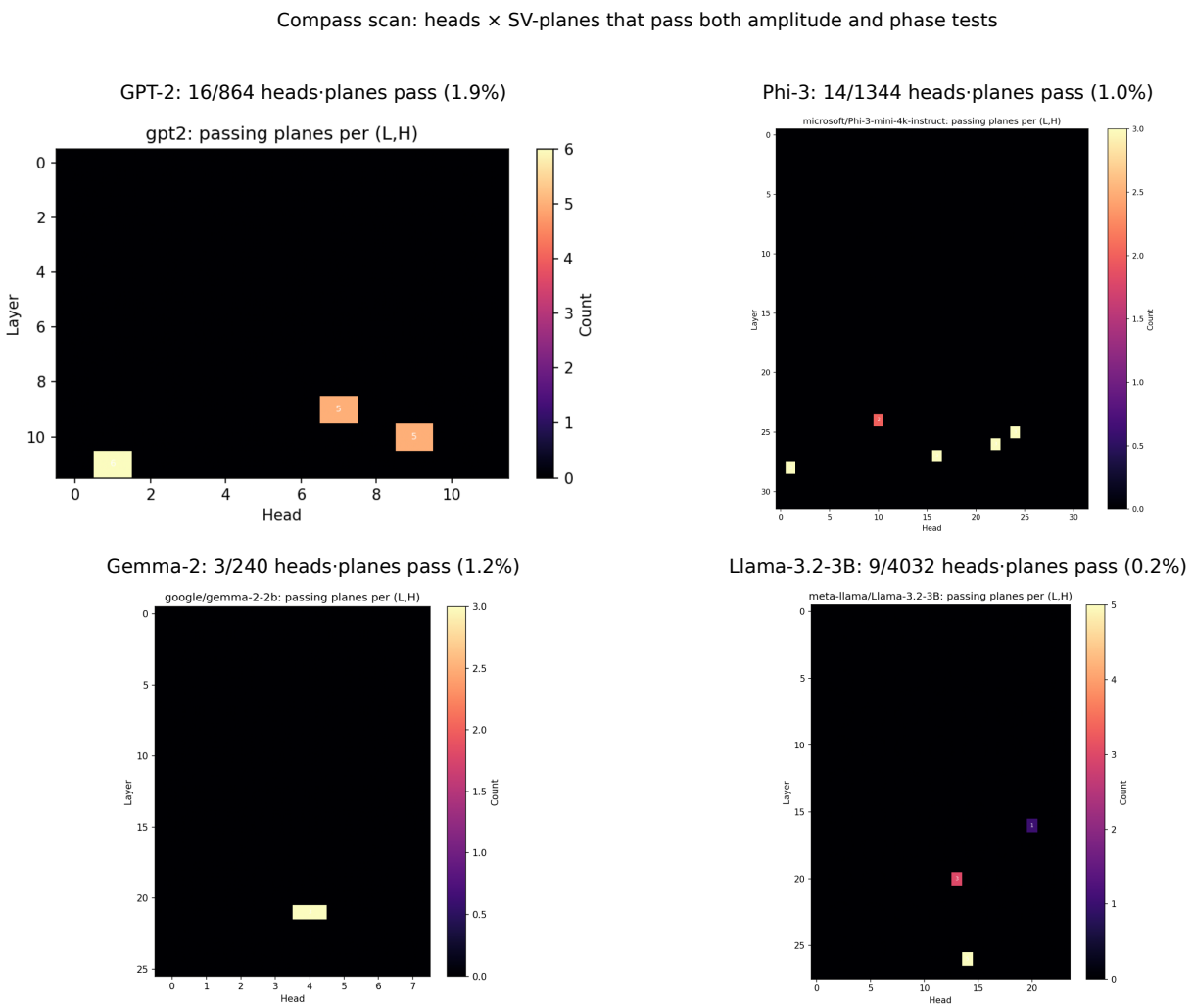

Figure 1: **Compass scan across four models.** Heatmap cells are `(layer, head)` pairs; bright cells mark heads whose top-4 SV planes pass both amplitude and phase tests. In every model, passing planes are rare (0.2–1.9%) and cluster at specific layers; Llama-3.2-3B is the sparsest but has the highest-amplitude compass.

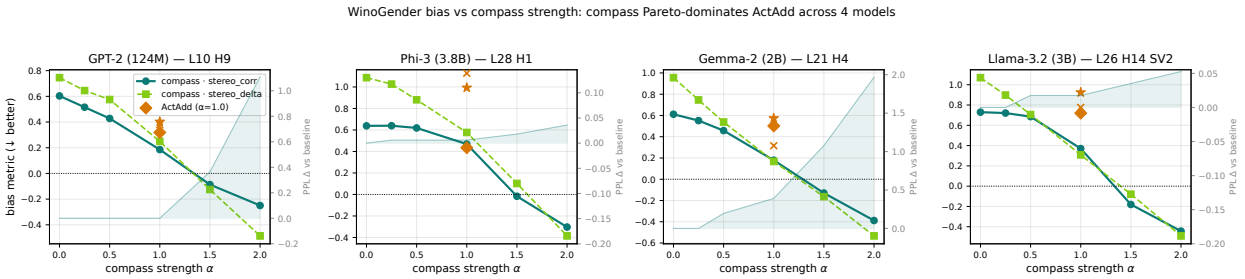

Figure 2: **WinoGender α-sweep across four models.** Solid teal: compass `stereo_corr`; dashed green: compass `stereo_delta`; orange diamond/star: ActAdd (α=1) anchor; shaded: PPL Δ vs baseline (right axis). In all four models, bias drops linearly with α, hits zero near α=1.5, and *reverses* at α=2 — strong causal evidence. PPL barely moves.

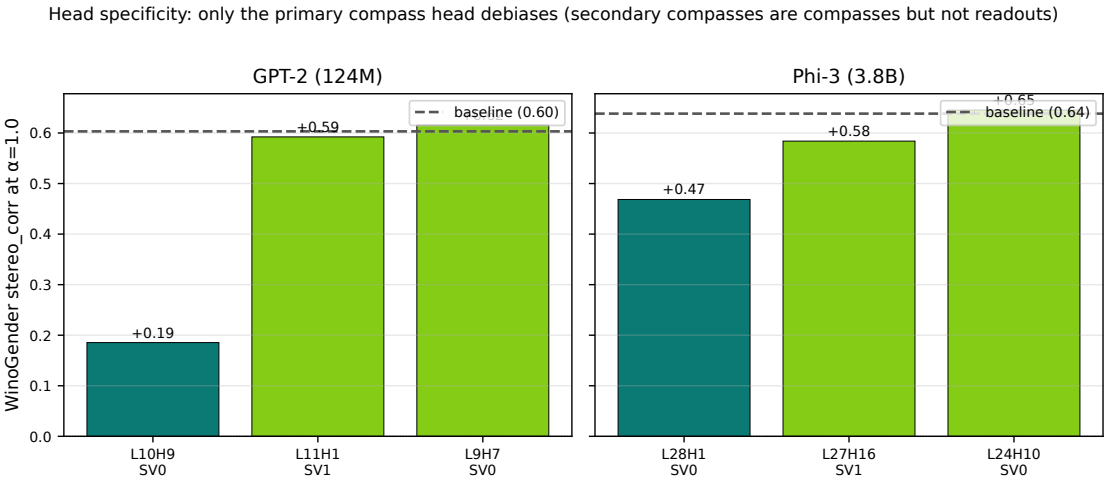

Figure 3: **Head specificity.** At α=1.0, only the *primary* compass head (dark teal) actually debiases; other passing compass heads in the same model (light green) leave `stereo_corr` near baseline. The dissociation rules out cherry-picking and supports a specific functional role for L10 H9 / L28 H1.

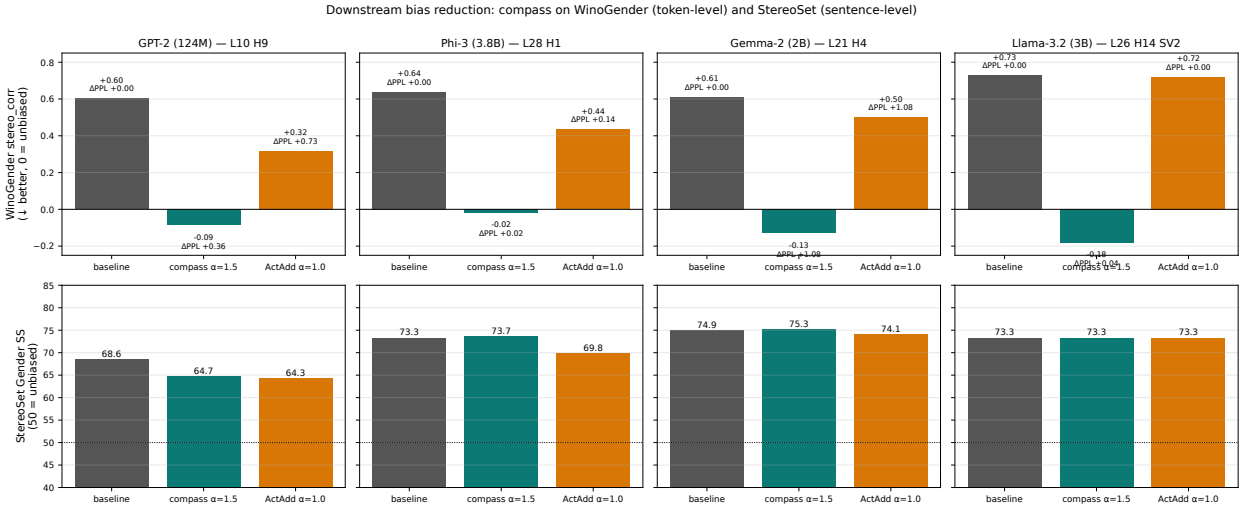

Figure 4: **Downstream bias reduction.** Top row: WinoGender `stereo_corr` (lower is better; zero is unbiased). Bottom row: StereoSet Gender SS (50 is unbiased). Compass works on all four models for WinoGender (token-level), but only GPT-2 for StereoSet (sentence-level).

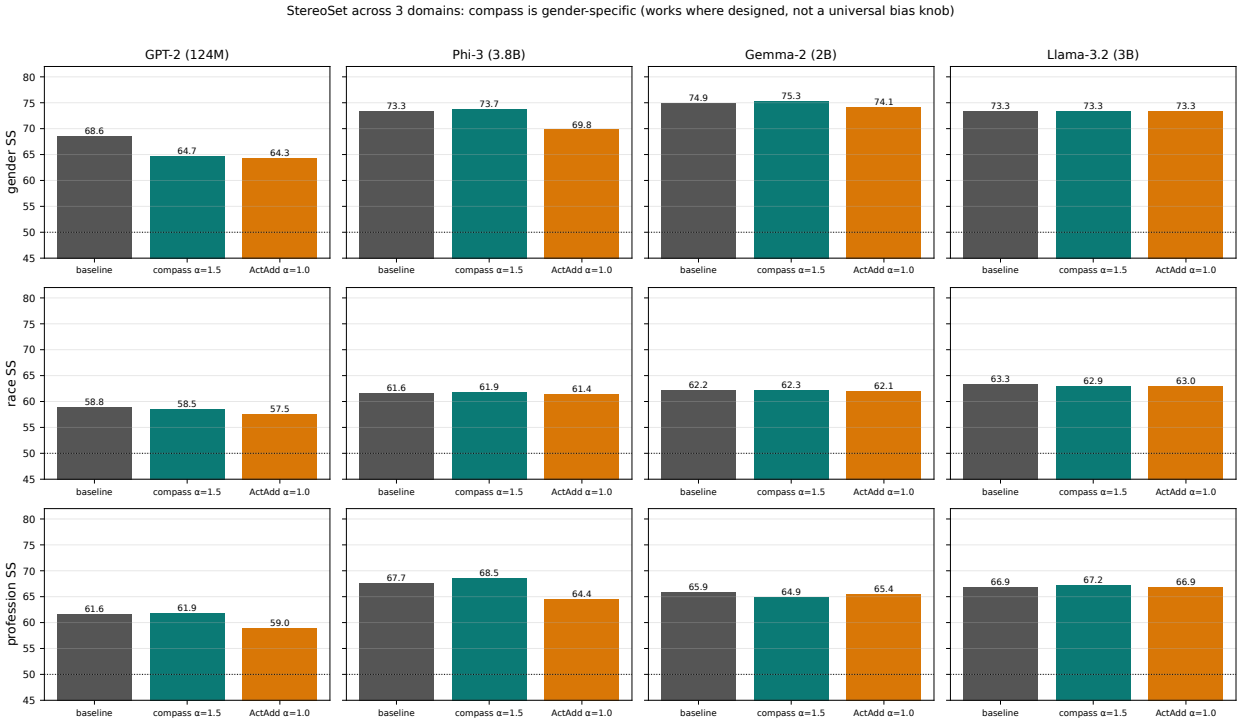

Figure 5: **StereoSet across three bias domains.** Compass barely moves race or profession SS; the *gender* compass is domain-specific and does not act as a universal bias knob. This is scientific evidence for specificity, not a bug.